

Board of Studies (BoS) Syllabus Template

Course Information

Degree: UG

Program: BCA

Course Name: Database Management Systems

Course Credits: 3

Semester:III

Course Category: Core

Type of Course: Lecture

New Course: No

Learning Hours

Activity	Hours per Week
Lecture	3
Tutorial	0
Practical	2
Experiential Learning	0

Course Contribution

Category	Percentage Contribution
Employment Enhancement	40%
Entrepreneurship Skills	30%
Skill Development	30%

Course Relevance

Level	Percentage Relevance
Global	30%
National	30%
Regional	20%
Local	20%

Course Outcomes (COs)

CO#	Course Outcome	Bloom's Taxonomy Level
C01	Describe the fundamental concepts of databases and utilize ER diagrams to design well-structured database systems.	BL-2
C02	Illustrate database design concepts, enforce keys and constraints, and use relational algebra for data manipulation.	BL-3
C03	Examine the SQL operations on Database.	BL-4
C04	Create views, indexing, normalization, and functional dependencies for efficient database design and management.	BL-6
C05	Utilize permissions, triggers, and transaction concepts with ACID properties to ensure effective and reliable database management.	BL-3

Module	Course Content	Tools Used / Assessment and Activity	Bloom's Level	Mapping COs	Percentage change in Module (put Nil if there is no change or it is a new course write NEW)
Module 1: Introduction to Databases	Definition of Data, Database, and DBMS, Overview of Database	PPT Presentation & Classroom interaction	BL-1	CO1	
	Applications, Advantages and Disadvantages of DBMS,		BL-2	CO1	
	Roles of Database Users and Administrators.		BL-4	CO1	
	Data Models: Introduction to Data Models, Types of Data Models		BL-2	CO1	
	Hierarchical, Network, Relational, Object-oriented.		BL-4	CO1	
	Importance of Data Models in DBMS		BL5	CO1	

	Entity-Relationship (ER) Model		BL-4	CO1	NIL
	Entities and Entity Sets, Attributes and Relationships		BL3	CO1	
	ER Diagrams,Key Constraints and Weak Entity Sets.		BL4	CO1	
Module 2: Database Design, Relational Algebra & Calculus	Introduction, Keys- Primary Key, Candidate Key, Super Key, Foreign Key		BL-2	CO2	
	Composite Key, Alternate Key, Unique Key NOT NULL, Check, Default		BL-4	CO2	
	Introduction to Relational Algebra		BL-2	CO2	
	Operations- Selection, Projection		BL-3	CO2	
	Set Operations		BL-3	CO2	
	Join Operations,Division		BL-3	CO2	
	Tuple and domain Relational Calculus		BL-4	CO2	
Module 3: Introduction to SQL	Introduction, Data Definition Language: Create, alter		BL-3	CO3	
	drop, rename and truncate statements		BL-3	CO3	
	Data Manipulation Language: Insert, Update and Delete Statements.		BL-3	CO3	
	Data Retrieval Language: Select statement.		BL-3	CO3	
	Transaction Control Language: Commit, Rollback and Savepoint statements.		BL-3	CO3	
	Single row functions using dual: Date, Numeric and Character functions.		BL-3	CO3	
	Group/Aggregate functions: count, max, min, avg and sum functions.		BL-3	CO3	
	Set Functions: Union, union all, intersect and minus. Subquery.		BL-3	CO3	
	Joins		BL-3	CO3	

Module 4: View & Normalizati on	Create view, types of view, and operations on views		BL-2	CO4	
	dictionaries for table, view, user, constraints		BL-4	CO4	
	creating view, Renaming the Column of a view, Creating Indexes		BL-3	CO4	
	Updating, Selection, destroying view		BL-3	CO4	
	Creating Indexes		BL-3	CO4	
	Normalization: Functional Dependencies: Definition, Properties,		BL-2	CO4	
	Types, Closure of Functional Dependencies,		BL-4	CO4	
	Normal Forms (1NF, 2NF)		BL-3	CO4	
	Normal Forms (3NF, BCNF)		BL-4	CO4	
Module 5: Data Control Language, Triggers, Transaction Concepts	Granting Permissions, Revoke permission		BL-2	CO5	
	Creating and managing User, Create, Role		BL-4	CO5	
	provide privileges, dictionaries		BL-3	CO5	
	Triggers syntax, Enabling/Disabling, Triggers, dictionaries		BL-2	CO5	
	Transaction Recovery – ACID Properties,		BL-3	CO5	
	Serializability, types of serializability - conflict serializability, view serializability.		BL-4	CO5	

Textbooks:	
1	Abraham Silberschatz, Henry F. Korth, S. Sudharshan, “Database System Concepts”, Fifth Edition, Tata McGraw Hill, 2016
2	RamezElmasri, Shamkant B. Navathe, “Fundamentals of Database Systems”, Fourth Edition, Pearson/Addision Wesley, 2017.

CO-PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
C01	3	2	1	2	1	1	2	1	1
C02	2	3	2	1	2	2	3	1	2
C03	2	3	3	2	2	3	3	2	3
C04	1	2	3	3	2	2	3	2	2
C05	2	3	2	3	3	3	3	3	3